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To cite this article: Rui Guo, Guijun Xian, Chenggao Li, Xiangyu Huang & Meiyin Xin (2021): Effect of fiber hybridization types on the mechanical properties of carbon/glass fiber reinforced polymer composite rod, Mechanics of Advanced Materials and Structures, DOI: [10.1080/15376494.2021.1974620](https://doi.org/10.1080/15376494.2021.1974620)

To link to this article: <https://doi.org/10.1080/15376494.2021.1974620>



Published online: 24 Sep 2021.



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ORIGINAL ARTICLE



Effect of fiber hybridization types on the mechanical properties of carbon/glass fiber reinforced polymer composite rod

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ABSTRACT

In order to balance the cost performance of carbon and glass fibers, the unidirectional carbon/glass hybrid fiber reinforced polymer composite rods were developed through the pultrusion technology in the present paper. Two kinds of fiber hybridization types were proposed including intra-yarn hybrid (uniform dispersed hybrid, UDH) and inter-layer hybrid (core-shell hybrid, CSH). Furthermore, the effects of fiber hybridization types on the mechanical properties (interface/short beam shear strength, three-point bending strength and tensile strength) were investigated experimentally. The full-field strain was obtained through the digital image correlation method to reveal the hybrid mechanism. It was found that the effects of fiber hybridization types on mechanical properties were remarkable. Through uniformly dispersing the carbon fiber into the glass fiber, the maximum increase percentages of available strength were up to 10.9% for short beam shear, 60.3% for three-point bending and 58.7% for tensile strength, respectively. The results of full-field strain analysis revealed that the mechanical failure of CSH rods was dominant by the interface debonding of shell/core, which significantly decreased the mechanical properties. In comparison, UDH rods had the intact interface bonding of carbon/glass fiber/resin, the mechanical failure was derived from the fracture of carbon fiber. The tensile strength of carbon fiber was fully utilized, which contributed to the higher mechanical properties. The larger scale carbon fiber tow may form the weaker interface layer of carbon/glass fiber/resin, which led to uneven interface stress distribution under the external loading and the decrease of interface shear strength.

ARTICLE HISTORY

Received 11 August 2021

Accepted 26 August 2021

KEYWORDS

Carbon/glass fiber hybrid rods; mechanical properties; digital image correlation; hybrid mechanism

1. Introduction

Fiber reinforced polymer (FRP) composites have been widely used in the fields of aircraft, ships, automobiles and wind turbine blades, etc. [1, 2]. Recently, FRP composites become the attractive materials in the field of civil engineering owing to the high ratio of strength-to-weight, and the high ratio of stiffness-to-weight, superior fatigue and corrosion resistances [3]. One of the most advanced technologies for fabricating the FRP products such as sheets, bars and rods can be pultrusion technology [4]. For FRP sheets, they are mainly used to strengthen the damaged concrete [5] or steel structures [6, 7]. For FRP bars, they are commonly used to reinforce the concrete structures by replacing steel bars [8]. Furthermore, FRP rods are used as the cable to replace traditional steel cable in bridge engineering [9] or as prestressing tendons to strengthen the concrete structures [10]. In addition, FRP profiles can also be used for the bridge deck and roof structures, etc. [11]. In conclusion, it has been proved that FRP composites can form a satisfactory combined structure with concrete and steel materials [12, 13].

The structures in civil engineering have the characteristics of long service life requirement (>50 years), harsh service environment (hydrothermal exposure, cyclic loading and ultraviolet aging etc.) and large demand for materials, which are the key and considerable factors in the process of promoting the large-scale engineering applications of FRP composites. Carbon fiber reinforced polymer (CFRP) composites possess excellent mechanical properties (high strength and modulus), superior fatigue and corrosion resistances [14]. However, the disadvantages of high price and lower elongation limit its large-scale applications in civil engineering. By contrast, glass fiber reinforced polymer (GFRP) composites possess higher elongation and low price, and they have been used as the structure materials in various fields. Regrettably, the poor fatigue and corrosion resistances restrict the applications in harsh civil engineering conditions [15, 16]. In order to make full use of material properties and reduce the production cost of FRP composites, the fiber hybrid method is seen as a promising strategy to achieve the optimal cost performance and has gained more attentions [17, 18].

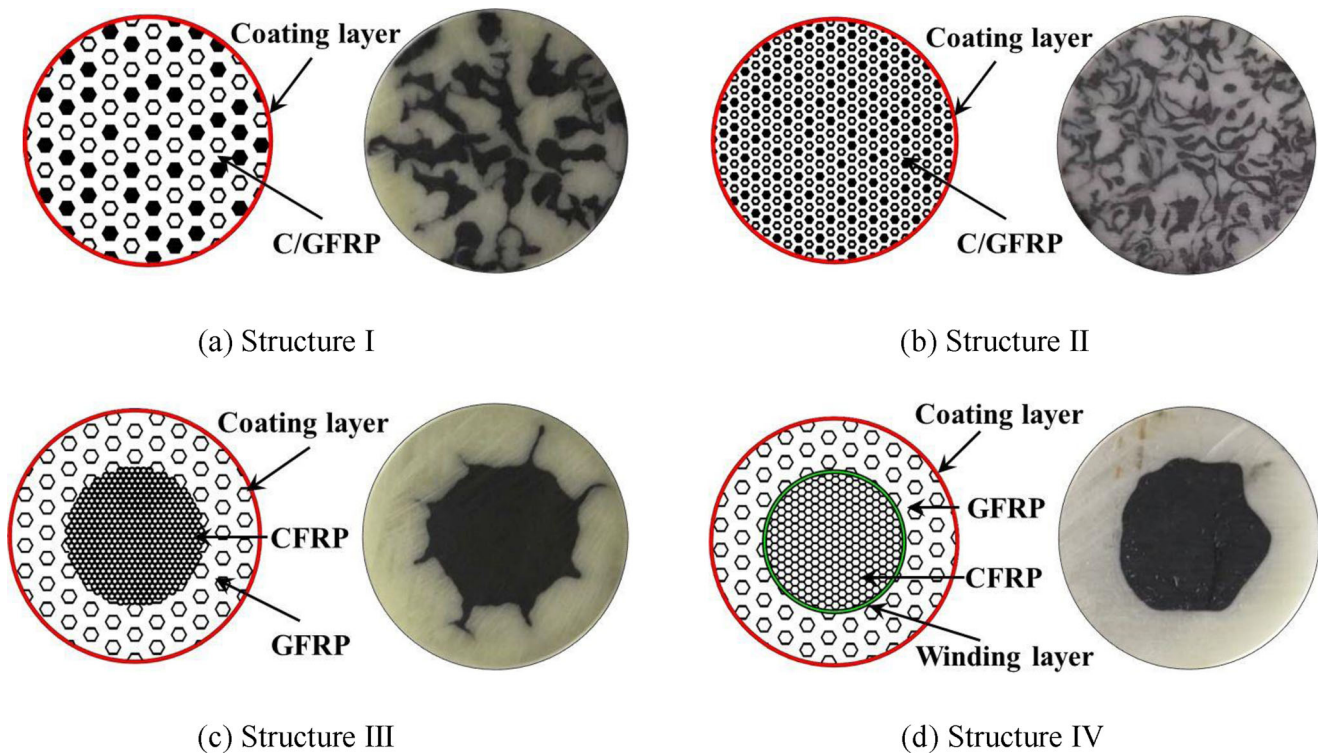


Figure 1. The cross-sections of four kinds of hybrid rods of (a) structure I, (b) structure II, (c) structure III and (d) structure IV.

In order to balance the price and properties of glass fiber and carbon fiber, the fiber hybrid technique was proved to be an effective method [19–21]. Related studies have focused on the effect of fiber hybridization types on mechanical properties of hybrid composites. Current fiber hybrid configurations mainly included carbon/glass fiber interlayer hybrid, intra-layer hybrid, intra-yarn hybrid and combined hybrid etc. [22]. Interlayer is the simplest and cheapest method through stack carbon and glass fiber onto each other, these type are sandwich structure and layer by layer hybrid configurations [23]. For intra-layer hybrid, the two fiber types are mixed within the layers. In addition, Carbon and glass fiber can also be mixed or co-mingled on the fiber level to form an intra-yarn hybrid. The positive hybrid effect was achieved through improving the failure strain of fiber with the low elongation (LE) [24]. For example, You et al. [25] found the failure strain of carbon/glass fiber hybrid rod with uniform fiber dispersed mode increased by 33% compared to the single CFRP rod. Wu et al. [26] studied the tension-tension fatigue properties of carbon/glass and carbon/basalt fiber hybrid plates, and found that the fatigue life of carbon/basalt fiber hybrid composites was more than that of carbon/glass fiber hybrid composites, which was because the interface bonding property of basalt fiber/resin was more superior compared to the glass fiber/resin. Xian et al. [27] investigated the tension-tension fatigue behavior of carbon/glass fiber hybrid rods, the results showed that the interface debonding of GFRP shell/CFRP core caused the splitting failure of GFRP shell, which brought about the premature fatigue failure of hybrid rods. In addition, through the fiber hybrid, the hybrid rod can also achieve a controlled and gradual failure mode (known as “pseudo-ductility”) [28–30]. That was because when carbon/glass fiber hybrid

composites were subjected to the tensile loads, the carbon fiber has an early breakage before the glass fiber owing to the low elongation, which was helpful to provide the warning sign before the final failure of materials and structures [31]. Jones et al. [32] found the pseudo-ductility behavior of hybrid composites was achieved by uniformly dispersing carbon fibers into glass fibers. Young et al. [25] did not observe the distinct pseudo-ductility behavior of hybrid rods owing to the volume fraction of carbon fiber was more than that of glass fiber. So far, the hybrid mechanism (such as the stress redistribution) of hybrid rods was still unclear and need the further investigation to obtain the optimal mechanical properties.

In the present paper, in order to balance the cost and properties, the unidirectional carbon/glass fiber hybrid rods with the diameter of 22mm were developed through the pultrusion technology. The mechanical properties of hybrid rods were investigated experimentally. The hybrid mechanism was revealed through the digital image correlation method, which will be conducive to the design, preparation and application of carbon/glass fiber hybrid rods.

2. Experimental procedures

2.1. Raw materials

In the present paper, the unidirectional carbon/glass fiber hybrid composite rods with the diameter of 22mm were developed by the pultrusion technology to apply to the bridges, offshore and petroleum engineering [33–35]. Compared to the rod with the small diameter, the large-diameter rod can realize the decrease of production cost. Meanwhile, the anchorage efficiency and durability can be

Table 1. The sizes of four kinds of hybrid rods, and the unit is mm.

Rod types	Diameter of CFRP core	Thickness of GFRP shell	Thickness of coating layer	Thickness of winding layer
Structure I	/	/	0.3	/
Structure II	/	/	0.3	/
Structure III	12	4.7	0.3	/
Structure IV	12	4.3	0.3	0.4

Notes: Coating layer was composed of surface mat, and winding layer was composed of glass fiber.

improved through reducing the rod number or exposure areas with the equivalent bearing capacity. There were two kinds of fiber hybrid configurations in the present research and they were named as intra-yarn hybrid (uniform dispersed hybrid, UDH) and interlayer hybrid (core-shell hybrid, CSH). Intra-yarn hybrid includes two kinds of cross-section morphology and was named as structure I and II, and inter-layer hybrid includes two types of cross-section morphology of structure III and IV. Furthermore, four types of hybrid rods were shown in Figure 1. Both structure I and II were composed of the external coating layer and internal carbon/glass fiber uniformly dispersed hybrid core. The carbon fibers in structure I was more uniformly dispersed than those in structure II. Both structure III and IV were composed of external coating layer, glass fiber reinforced polymer (GFRP) shell and internal carbon fiber reinforced polymer (CFRP) core. The difference between the both structures was that structure IV has an extra winding layer compared to structure III. The adopted width of coating layer (surface mat) was equal to the cross-section perimeter of hybrid rod. Its length was consistent with the fiber tow. In the pultrusion process, the surface mat was attached on the surface of the fiber tow after the rod preforming, and then the surface mat and fiber tow were pulled together into the curing mold and cured. In addition, the sizes of four kinds of hybrid rods were listed in Table 1. The fiber volume fractions of four types of hybrid rods were $\sim 70\%$, and the volume fraction of carbon fiber was $\sim 19\%$. 12 K carbon fibers (Shanghai Petrochemical Co., Ltd, Shanghai, China), glass fibers (OC 2400tex, Owens corning, America) and the bisphenol-A epoxy matrix (Airstone TM 1122E, Blue Cube Chemicals Company, Zhangjiagang, China) were adopted in the present study, the hardener was anhydride type and its ratio to resin was 1:1.14. The curing process included the initial curing in the mold and the posturing in the oven. The mold was divided into three curing zones with the temperature of 147°C , 195°C and 195°C , respectively, and the length of each curing zone was 250 mm. The oven also included three posturing zones with the temperature of 195°C , 195°C and 147°C , respectively, and the length of each curing zone was 100 mm. In addition, the mechanical properties of above mentioned fibers and resin were listed in Table 2.

2.2. Mechanical tests

2.2.1. Shear tests

In the present paper, two types of shear tests were conducted to characterize the interface bonding properties between the fiber and resin, including the short beam shear strength (SBSS) and interface shear strength (ISS) tests.

According to ASTM D4475, the short beam shear samples were cut into 132 mm in length with a span of 110 mm as shown in Figure 2. The loading speed of the universal tensile machine (DHY-10080, Shanghai, China) was set to 1.6 mm/min. Five samples were repeated to obtain the average, and the full field strain was monitored by the digital image correlation method to obtain the strain distribution.

As shown in Figure 1, UDH rod has a uniformly dispersed interface of carbon/glass fiber/resin. In comparison, CSH rod has the special GFRP shell/CFRP core interface, and the diameter of CFRP core was 12 mm. In order to obtain the bonding property of above two interfaces, a new testing device was developed as shown in Figure 3, including the pressure lever, fixed roof and bearing pedestal. The pressure lever with the diameter of 12 mm was used to test ISS of hybrid rods based on the diameter of CFRP core for the core-shell hybrid rod. The samples were cut into cylinder with the height of 5 mm. Actual dimensions of each specimen were measured by micrometer with an accuracy of ± 0.001 mm. The loading speed of the universal tensile machine (DHY-10080, Shanghai, China) was set to 1.0 mm/min. Five samples were repeated to obtain the average.

2.2.2. Three-point bending strength (TPBS) tests

The three-point bending samples were cut into 216 mm in length with a span of 180 mm as shown in Figure 4. According to ASTM D4476, the suggested ratio of bending specimen thickness to rod diameter is from 0.146 to 0.5. Therefore, the specimen thickness in the present paper was determined to be 10.5 mm with a thickness to diameter ratio of 0.477. Actual dimensions of each specimen were measured with a micrometer of an accuracy ± 0.001 mm. The crosshead displacement rate of the universal tensile machine (DHY-10080, Shanghai, China) was set to 6 mm/min. Five samples were repeated to obtain the average.

2.2.3. Tensile tests

An extrusion-bonding compound anchorage was adopted to anchor the hybrid rods for the tensile tests, and the anchorage system included the steel tube, steel wedge, filler (epoxy) and central ring [27]. The tensile samples were cut into 800 mm in length with one free section of 200 mm in length and two anchor sections of 300 mm in length, as shown in Figure 5(a). The loading speed of the universal tensile

Table 2. Mechanical properties of fibers and resin.

Mechanical properties	Carbon fiber	Glass fiber	Resin
Tensile modulus (GPa)	235	90	2.66
Tensile strength (MPa)	3810	2604	109
Elongation (%)	1.62	2.90	4.20

machine (WAW-1000, Shanghai, China) was set to 2 mm/min. Five samples were repeated to obtain the average. In addition, the full field strain and the failure modes were obtained by digital image correlation method.

To investigate the hybrid effect, CFRP, GFRP and C/GFRP hybrid plates were prepared from CFRP core, GFRP shell and UDH rod with the dimension 250 mm $\times$ 15 mm $\times$ 1.0 mm to conduct the tensile tests according to ASTM D3039, as shown in Figure 5(b). Actual dimensions of each specimen were measured by the micrometer with an accuracy of ± 0.001 mm. The loading speed of the universal tensile machine (DHY-10080, Shanghai, China) was set to 2 mm/min. Five samples were repeated to obtain the average.

2.3. Full field strain monitoring

The full field strains of hybrid rods during the short beam shear and tensile loading were obtained through the digital image correlation method. The surfaces of hybrid rods were sprayed with the white paint as the background, and then the black paint with high contrast speckle pattern was sprayed on the surface of white background. It was noted that the size and distribution of speckle should be controlled as evenly as possible to obtain the better speckle quality. After this, the samples were dried at room temperature to conduct the mechanical tests. Two white light sources were utilized to illuminate the specimen surface. Speckle and calibration plate images were taken by using a pair of point

gray grasshopper five megapixel cameras fitted with 35 mm lenses with the rate of one frame per second. Finally, the speckle and calibration plate images were imported into the post-processing software (VIC-3D8, Correlated Solutions, Inc, America) to obtain the full field strain.

3. Results and discussion

3.1. Shear properties

The shear strength can be used to characterize the bonding performance between fiber and resin [36–38]. Two kinds of shear properties of hybrid rods were obtained, including the short beam shear strength and interface shear strength as shown in Table 3. The load-displacement curves of short beam shear were shown in Figure 6(a). The load-displacement curves of structure I, II and III showed a typical elastic characteristic. Furthermore, the failure modes were found to be abrupt and catastrophic shear failure at the neutral axis of cross-section without warning. There was no inflection point appearing in the load-displacement curves until the load reached the ultimate value. In comparison, it can be seen that the rod of structure IV presented a two-stage shear behavior. In the first stage (0 to 23 kN), the load-displacement curve was similar with the above three kinds of rods. The second stage was controlled by the debonding failure of shell/core interface, which led to an obvious inflection point in the load-displacement curve. Meanwhile, larger breakage ductility was also detected compared to the other three



Figure 2. Short beam shear testing device diagram of hybrid rods.

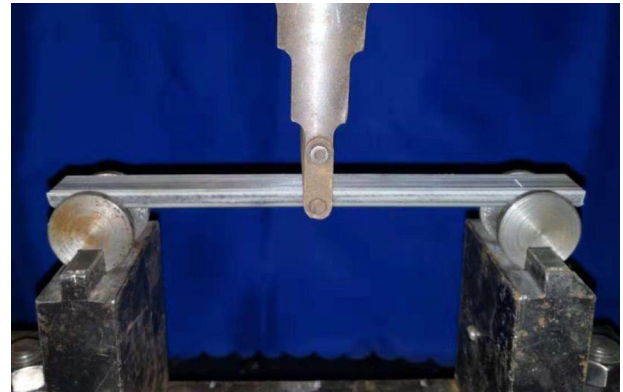


Figure 4. Three-point bending testing device diagram of hybrid rods.

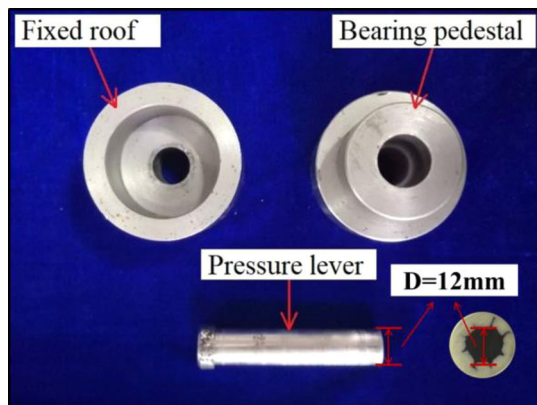
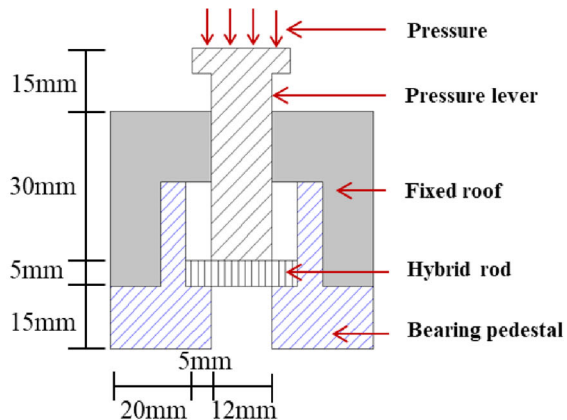


Figure 3. Interface shear testing device diagram of hybrid rods.





(a) Anchorage system of hybrid rod



(b) Anchorage system of hybrid plate

Figure 5. The tensile samples of (a) hybrid rods and (b) hybrid plates.**Table 3.** SBSS and ISS of hybrid rods.

Rod types	SBSS (MPa)	ISS (MPa)
Structure I	61.72 (± 0.24)	58.29 (± 1.78)
Structure II	60.85 (± 2.22)	63.29 (± 0.91)
Structure III	55.65 (± 1.49)	56.40 (± 4.35)
Structure IV	59.66 (± 0.23)	43.12 (± 2.66)

kinds of rods. SBSS were calculated according to ASTM D4475 and listed in Table 3. It can be seen that the obvious hybrid effect was realized through uniformly dispersing the carbon fiber into the glass fiber, which brought about a maximum increase percentage (10.9%) of SBSS for the UDH rod compared to the CSH rod. In addition, it can be found that the effect of the dispersity (bundle size) of carbon fiber in the rod on SBSS was negligible, which contributed to the similar SBSS between structure I and structure II.

The full-field strain of short beam shear under the ultimate load was obtained by VIC-3D as shown in Figure 7. As seen, the ultimate strains at mid-span of structure I, structure II, structure III and structure IV were 1.22%, 1.24%, 1.23% and 1.48%, respectively. Larger ultimate strain at mid-span of structure IV was attributed to the interface debonding of shell/core led to the stress redistribution from CFRP core to GFRP shell [27]. In addition, the interface debonding of shell/core of structure IV resulted in the decrease of stiffness, which prevented the extension of stress concentration in mid-span to the CFRP core and decreased the peak of stress concentration zone (red part).

In order to further reveal the failure mechanism of short beam shear, 200 points were selected at equal distance from top to bottom along the specimen thickness of the quarter span to plot the stain distribution as shown in Figure 8. As shown, the strain distributions of structure I and structure II were linear, and the line intersection at different loading levels did not move down. This proved that there was no interface debonding of shell/core, which was consistent with the failure mode (Figure 6b). As for structure III and structure IV, it can be observed that the line intersection (neutral axis) moved down with the increase of shear load. This was because the interface debonding of shell/core led to the

stress redistribution from CFRP core to GFRP shell on the tension side, which brought about the increase of the tensile stress of bottom GFRP shell. Therefore, the neutral axis must be moved down to ensure that the bending moments produced by the tensile force and compressive force are equal. It should be noticed that the interface debonding of shell/core of structure IV was more serious than that of structure III, which can also be demonstrated by the variation of ISS and TPBS as following.

ISS of four kinds of hybrid rods were shown in Table 3. Through uniformly dispersing the carbon fiber into the glass fiber, the obvious hybrid effect brought about a maximum increase percentage (46.78%) of ISS for the UDH rod compared to the CSH rod. ISS of structure I was lower than that of structure II, which indicated that the effect of dispersity of carbon fiber tow on ISS was remarkable. Although the same materials were used in structure I and II and the interface bonding performance of fiber and resin were same in above two structures, the larger carbon fiber tow in structure I may lead to the stress concentration between carbon and glass fiber tows under the external loading, which significantly reduced ISS of structure I. Compared with structure III, the extra winding layer in structure IV made the GFRP shell separate from the CFRP core completely, which contributed to the decrease of ISS. In general, the fiber hybridization types of hybrid rod had an obvious effect on ISS. Uniform fiber dispersing mode [39] was considered as the desirable hybrid technology with the excellent shear properties.

3.2. Bending properties

The stress-strain curves of three-point bending of hybrid rods were shown in Figure 9. The details of three-point bending strength and modulus were shown in Table 4. It was obvious that the stress-strain curves of structure I and II were almost linear during the bending loading. Through the uniform fiber dispersion, the maximum increase percentage of three-point bending strength was observed up to

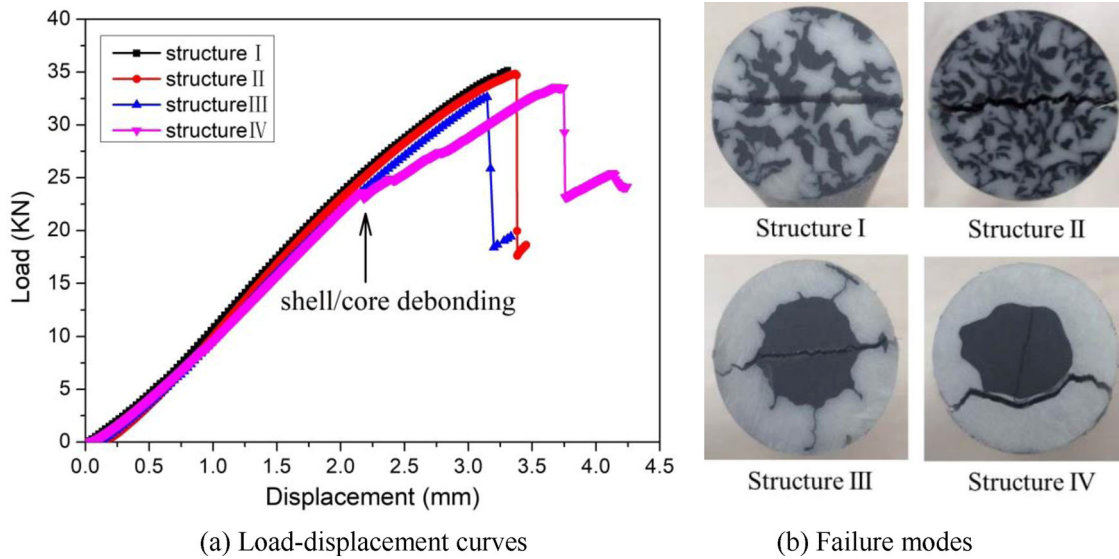


Figure 6. The short beam shear properties of hybrid rod of (a) load-displacement curves and (b) failure modes.

60.27% compared to structure III and IV. Meanwhile, the increase percentage of three-point bending modulus was up to 39.60% compared to structure III and IV. Furthermore, it can be observed that the effect of the dispersity of carbon fiber tow in the rod on TPBS was insignificant, which was similar with SBSS. In comparison, a two-stage stress-strain behavior was observed for the structure III and structure IV. The first stage was linear with the constant bending stiffness, and the obvious interface debonding of shell/core (inflection point as shown in Figure 9) was detected in the second stage. Furthermore, it can be found the shell/core interface debonding of structure IV was more serious than that of structure III, which was verified by an early arrival of inflection point. Based on this phenomenon, it can be concluded that the available bending strength before the interface debonding of shell/core for structure III and structure IV were 1243.8 MPa and 921.2 MPa, respectively.

The bending fracture failure mode of UDH rod occurred in carbon fiber on the outer surface. This was because the interface bonding of carbon/glass fiber/resin was intact during the bending loading process, and the local strains of carbon fiber and glass fiber were same. With the increase of bending load, the local strain of UDH rod reached the ultimate strain of carbon fiber, leading to the breakage of carbon fiber and the stress redistribution from the carbon fiber to glass fiber. When the glass fiber was unable to bear the residual bending load, the premature fracture failure of rod occurred. In comparison, the failure mode of CSH rod derived from the interface debonding of shell/core and ended up with the fracture of GFRP shell. In the initial loading stage, the interface bonding of shell/core was intact. The stress-strain curves of rod presented on the similar linear variation with the UDH rod. With the increase of bending load, the interface debonding of shell/core and the stress redistribution from CFRP core to GFRP shell occurred owing to the stiffness difference of CFRP core and GFRP shell. With the further deterioration of interface debonding, CFRP core completely lost the bearing capacity and the bending load was transferred to GFRP shell until the

catastrophic fracture of GFRP shell. In summary, the uniform fiber dispersion of hybrid rod can realize the significant increase of three-point bending strength and modulus through avoiding the premature interface debonding existing in the shell/core hybrid rod.

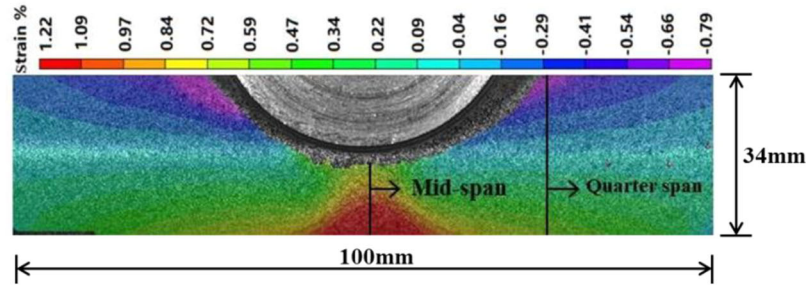
3.3. Tensile properties

In the present paper, the hybridization types of rods included the uniform dispersed hybrid and core-shell hybrid. To investigate the effect of fiber hybridization types on them hybrid effect [40, 41], two kinds of representative rods were selected including structure II and IV (Table 1). Furthermore, the rule of mixture was adopted to predict the tensile response of above two kinds of hybrid rods. For the rod of structure II, a perfect fiber/resin interface bonding was assumed, and the tensile strengths of hybrid rod at the elastic and ultimate rupture stages were calculated as following [42]:

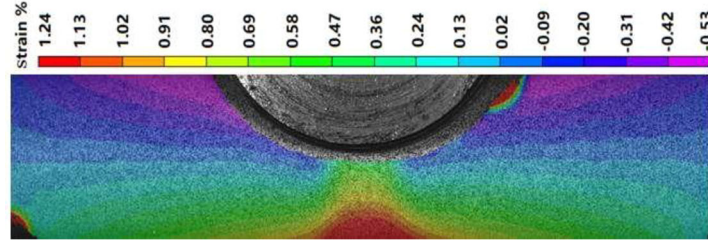
$$\sigma_{H,1} = (E_c V_c + E_g V_g + E_m V_m) \varepsilon_c \quad (1)$$

where $\sigma_{H,1}$ is the tensile strength in the elastic; ε_c is the failure strains of carbon fiber; V_c , V_g and V_m are the volume fraction of carbon fiber, glass fiber and resin, respectively.

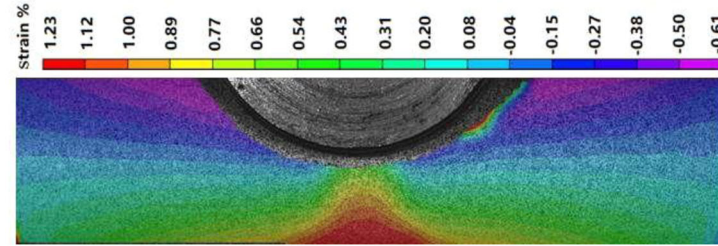
For the rod of structure IV, the assumptions of perfect interface bonding of fiber/resin in the elastic stage and poor interface bonding of shell/core in the ultimate rupture stage were made. CFRP core can continue to bear the load (no breakage occurred) through the interface debonding of shell/core when the strain reached ε_c , because the load of CFRP core can be transferred to GFRP shell. However, the real strain of CFRP core may be lower than ε_c , and the specific value was dependent on the degree of interface debonding. The more serious debonding led to the smaller strain of CFRP core. Therefore, it was difficult to capture the real strain of CFRP core. Based on this, it was supposed that the local strain of CFRP core was equal to ε_c regardless of the interface debonding degree. Furthermore, the tensile



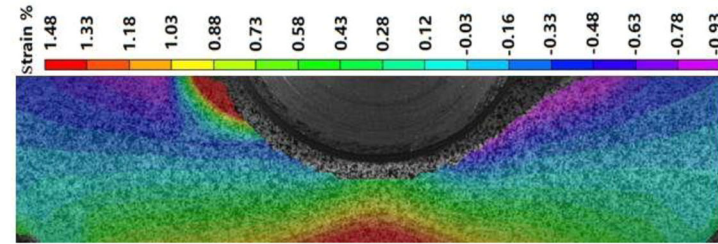
(a) Structure I



(b) Structure II



(c) Structure III



(d) Structure IV

Figure 7. Full-field strain of short beam shear under the ultimate load of (a) structure I, (b) structure II, (c) structure III and (d) structure IV.

strengths of rod for the structure IV at the elastic stage and ultimate rupture stages were calculated as following:

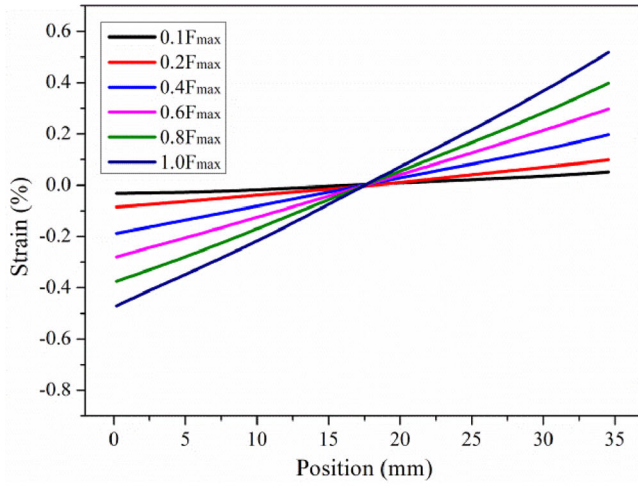
$$\begin{aligned}\sigma_{H,1} &= (E_c V_c + E_g V_g + E_m V_m) \varepsilon_c \\ \sigma_{H,2} &= (E_c V_c + E_g V_g + E_m V_m) \varepsilon_g\end{aligned}\quad (2)$$

where $\sigma_{H,2}$ is the tensile strength in the ultimate rupture stages; ε_g is the failure strains of glass fiber.

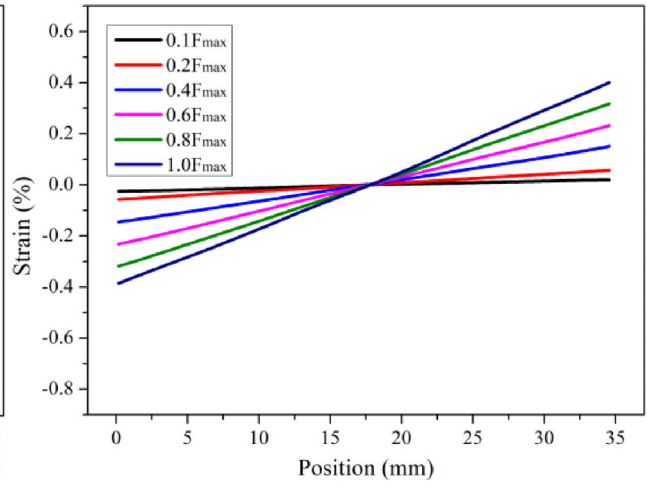
The effect of fiber hybridization types on the tensile response was investigated through the tensile tests of rod and plate. It was noted that the plates were adopted to study the hybrid effect of rod. The tensile properties of hybrid

rods were shown in Figure 10, and the detailed tensile results were listed in Table 5. It can be found that the excellent hybrid effect was realized by the uniform dispersion of carbon fiber into the glass fiber. Compared to the structure IV, the increase percentage of tensile strength of rod for the structure II was up to 58.69%. In addition, it can be found that the tensile modulus of two kinds of rods was almost owing to the same fiber volume fraction.

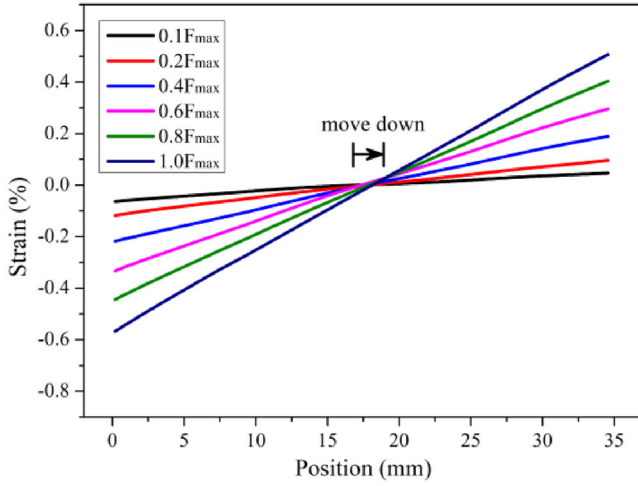
As seen in Figure 10(a), the stress-strain curve of structure II had only one elastic stage with a brittle failure mode. Furthermore, the predicted result according to Eq. (1) had



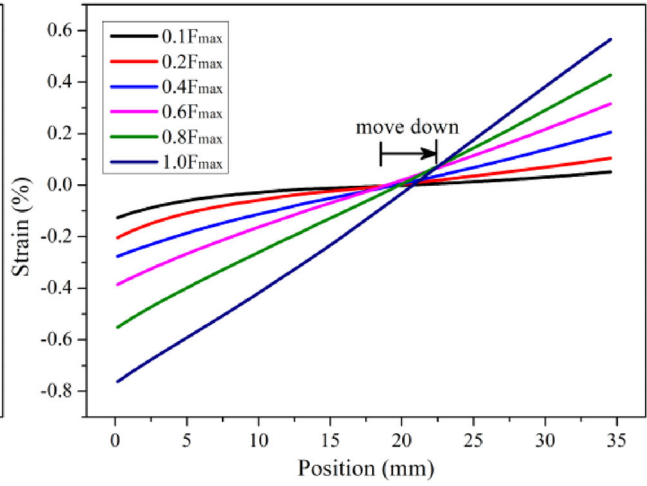
(a) Structure I



(b) Structure II



(c) Structure III



(d) Structure IV

Figure 8. The variation of strain distribution with the short beam shear load of (a) structure I, (b) structure II, (c) structure III and (d) structure IV.

the deviation with the tensile data of structure II. It can be found that $\sigma_{H,1}$ was less than the predicted value (point A), which was because the uneven stress redistribution from carbon fiber to glass fiber after the failure of carbon fiber resulted in the premature failure of rod for the structure II [43]. By contrast, the stress-strain curve of structure IV presented a two-stage behavior, including the initial elastic stage and the subsequent core-shell interface debonding stage. In the first stage (O→B), the tensile stiffness of rod was similar with the structure II, and the stress-strain curve was consistent with the predicted result from Eq. (2). Subsequently, the interface debonding of shell/core caused the decrease of tensile stiffness until the final failure of GFRP shell. Furthermore, the tensile strength obtained was lower than that of predicted value in this stage, which was because the actual strain of CFRP core was lower than ε_c until the tensile failure. The ultimate stage of structure IV had no load descent stage (Eq. (1)), which further verified the assumption of Eq. (2). It should be noticed that the available tensile strength of rod for the structure IV was 773.1 MPa (point B)

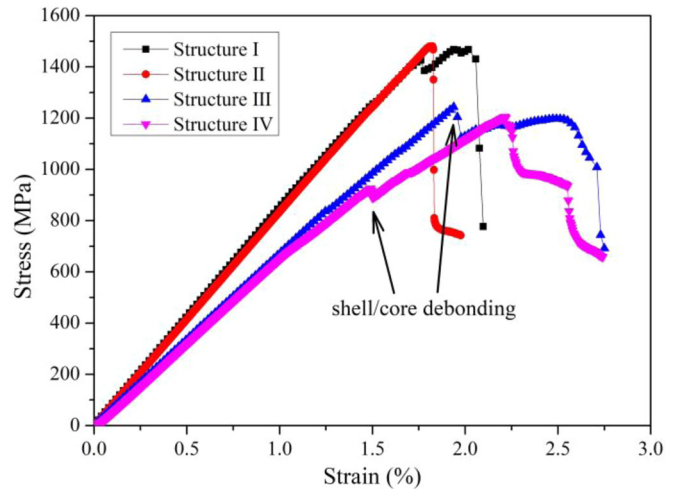


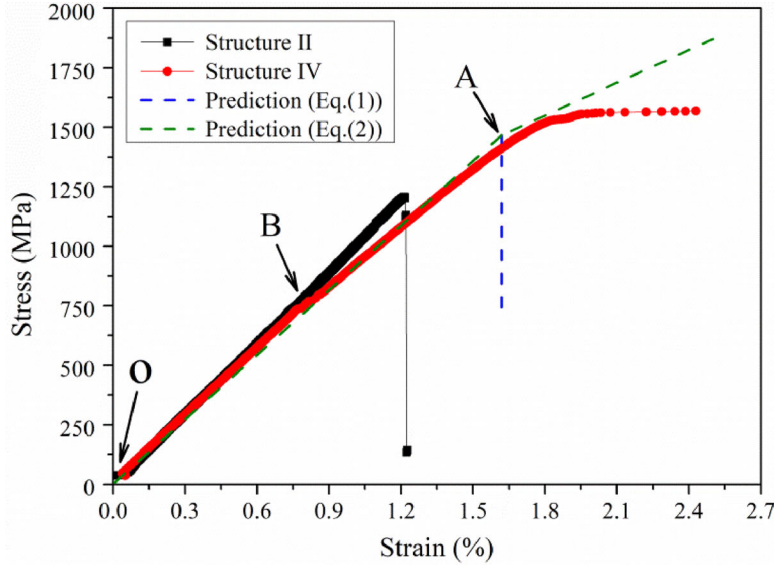
Figure 9. The stress-strain curves of three-point bending of hybrid rods.

in the first stage. This was because the initial interface debonding of shell/core occurred in point B, which undoubtedly accelerated the formation and propagation of

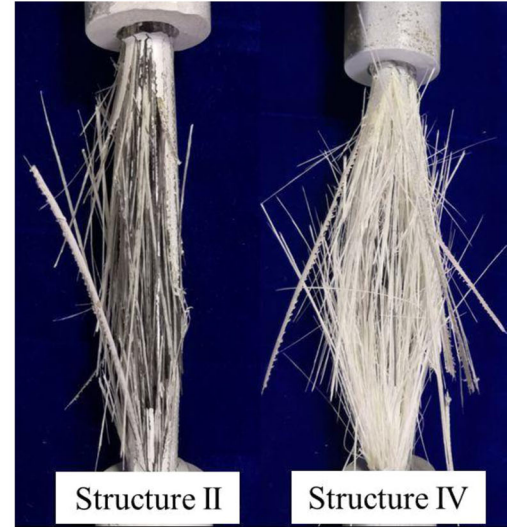
Table 4. Three-point bending strength and modulus of hybrid rods.

Bending properties	Structure I	Structure II	Structure III	Structure IV
Bending strength (MPa)	1476.4 (± 71.1)	1475.8 (± 42.3)	1243.8 (± 67.6)*	921.2 (± 23.4)*
Bending modulus (GPa)	54.32 (± 2.01)	54.11 (± 0.31)	46.49 (± 0.85)	38.91 (± 0.67)

Note: The data with the mark of "*" represented the available bending strength before the interface debonding of shell/core for structure III and IV.



(a) Stress-strain curves



(b) Failure modes

Figure 10. The tensile properties of hybrid rods of (a) stress-strain curves and (b) failure modes.**Table 5.** The tensile properties of hybrid rods.

Rod types	Tensile strength (MPa)	Tensile modulus (GPa)	Elongationb (%)
Structure II	1226.9 (± 57.6)	95.55 (± 3.68)	1.28 (± 0.04)
Structure IV	773.1* (± 62.8)	95.18 (± 3.89)	0.81* (± 0.08)

Notes: The data with the mark of "*" represented the available tensile strength and elongation before the interface debonding of shell/core for the rod of structure IV.

fatigue microcracks [27]. Therefore, the tensile strength of point B was regarded as the critical strength index when considering the actual service condition (cyclic loading and harsh environment). As seen in Figure 10(b), the failure mode of rod for the structure II was the burst failure of glass fiber and carbon fiber. In contrast, the burst failure of GFRP shell was the main failure mode for structure IV.

Based on the above discussion, the effect mechanism of fiber hybridization types on the tensile properties of two kinds of hybrid rods was summarized in Figure 11. As shown, the interface of carbon/glass fiber/resin of structure II had no debonding during the whole loading process, and the local strain of carbon fiber in the cross-section was the same as the glass fiber. With the increase of tensile load, the breakage of carbon fiber occurred when the strain reached its elongation. Subsequently, the uneven stress redistribution from carbon fiber to glass fiber led to the stress concentration of glass fiber. Finally, the catastrophic failure of rod for the structure II occurred when the residual tensile load surpassed the ultimate bearing capacity of glass fiber. For the rod of structure IV, the intact interface bonding property of shell/core in the elastic stage brought about the similar

variation of stress-strain curve with the rod of structure II. Owing to the glass fiber winding layer had angle difference with the longitudinal of rod, and the fiber fraction was lower in winding layer compare to CFRP core and GFRP shell. Therefore, the structure IV had a physical defect in winding layer. In addition, the physical and mechanical difference between CFRP core and GFRP shell was larger than structure II, which brought about the serious stress concentration in the core-shell interface under the external loading. With the increase of tensile load, the interface debonding of shell/core subsequently occurred and the slope of stress-strain curve decreased. Furthermore, the tensile load of CFRP core was transferred to GFRP shell, leading to the strain increase of GFRP shell. When the strain of GFRP shell reached its elongation, the catastrophic failure of rod for the structure IV was observed (Figure 10b).

The full-field strain distribution of two kinds of hybrid rods for the structure II and structure IV under the ultimate load was obtained by VIC-3D and used to verify the above tensile failure mechanism. As shown in Figure 12, the half-length of the rod was given due to the symmetry, and the anchorage end was near the anchor and the free end was located in the middle of the rod. As seen in Figure 12(a), the carbon fiber tow (marked as purple) of structure II had a first breakage under the ultimate tensile load, and the corresponding failure morphology was given in speckle image. The strain of carbon fiber after the fracture did not decrease to zero owing to the constraint effect from the surrounding glass fiber. Meanwhile, the strains of the anchor end and

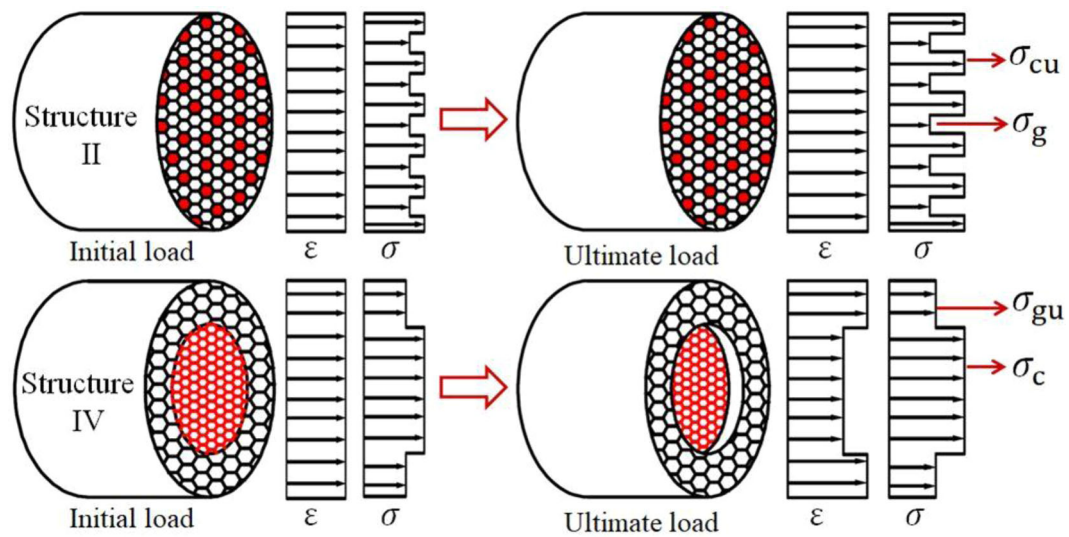
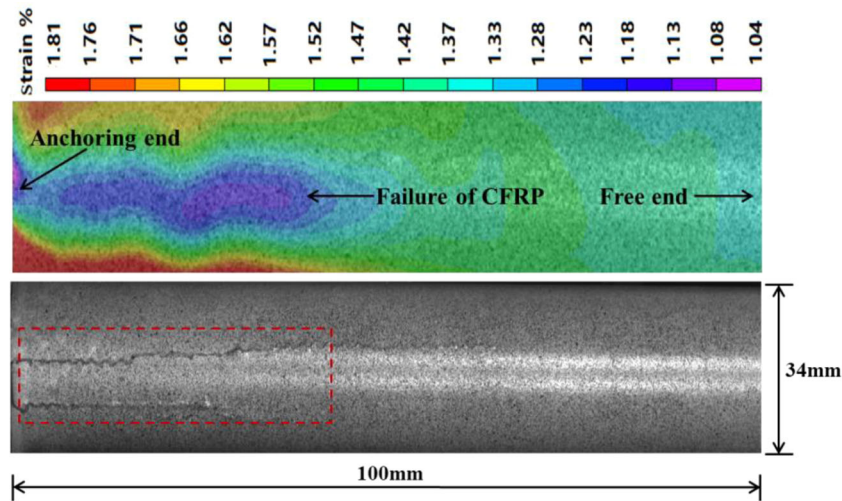
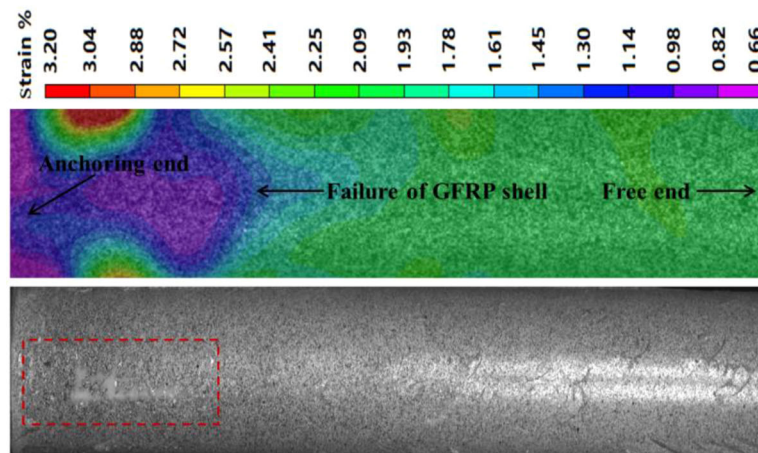


Figure 11. The tensile failure mechanism of rod for the structure II and IV. σ_{cu} and σ_{gu} represented the ultimate stress of carbon fiber and glass fiber.

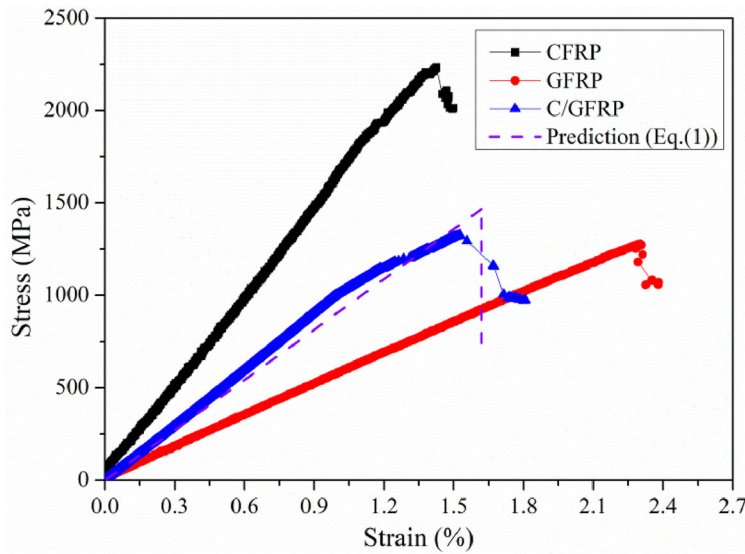


(a) Structure II



(b) Structure IV

Figure 12. Full-field strain distribution and speckle image of rod under ultimate tensile load of (a) structure II and (b) structure IV.



(a) Stress-strain curves



(b) Failure modes

Figure 13. The tensile properties of three kinds of plates of (a) stress-strain curves and (b) failure modes.

free end were 1.81% and 1.3%, respectively. The strain difference was because the uneven stress redistribution between the carbon fiber and glass fiber led to the stress concentration of glass fiber. Figure 12(b) shows the full-field strain and speckle image of structure IV. Similarly, the initial breakage of GFRP shell (marked as purple) was observed under the ultimate tensile load, and the corresponding failure morphology was also given in speckle image. It can be seen that the free end strain of structure IV was higher than that of structure II, which verified that the uniform stress redistribution from CFRP core to GFRP shell occurred after the interface debonding of shell/core.

After analyzing the hybrid mechanism of rod, the tensile properties of C/GFRP hybrid plate was also compared with the single CFRP and GFRP, as shown in Figure 13. As shown in Figure 13(a), the stress-strain curves of CFRP and GFRP plates were linear during the loading process. In comparison, the curve of C/GFRP plate presented a two-stage behavior including the first elastic stage and the second ductility stage. Furthermore, the tensile results of three kinds of plates were listed in Table 6. As shown, the tensile data of C/GFRP plate was consistent with the results of predicted model (Eq. (1)) in the elastic stage. Owing to the loading process was controlled by the displacement rate, the load carried by the specimen have a suddenly drop when the carbon fiber occur breakage, the glass fiber can continue bear the load due to the local strain is lower than the breakage strain of glass fiber. However, there was a short ductility stage and had no ascending phase of tensile strength, this was because the carbon fiber had the first breakage when the local strain reached ε_c , the uneven stress redistribution from carbon fiber to glass fiber caused the premature tensile burst failure of C/GFRP plate.

From the above results and discussion of tensile properties for the hybrid rods and plates, it can be concluded that the higher available tensile strength of rod for the structure II was obtained through the uniform fiber dispersion to

Table 6. Tensile properties of three kinds of plates.

Plate type	Tensile strength (MPa)	Tensile modulus (GPa)	Elongation (%)
CFRP	2400.8 (± 57.2)	171.80 (± 5.68)	1.39 (± 0.03)
GFRP	1235.0 (± 66.5)	58.92 (± 3.67)	2.35 (± 0.06)
C/GFRP	1283.9 (± 77.6)	94.82 (± 4.20)	1.35 (± 0.04)

avoid the interface debonding of shell/core compared to structure IV. Meanwhile, the effect of hybridization types on tensile modulus was negligible owing to the same fiber volume fraction. In addition, it can be observed that the tensile strength and modulus of hybrid rod (structure II) and hybrid plate were almost, which indicated the negligible effect of materials shape on the tensile properties.

4. Conclusions

In the present paper, two kinds of carbon/glass fiber hybrid rods were developed including intra-yarn hybrid (uniform dispersed hybrid, UDH) and inter-layer hybrid (core-shell hybrid, CSH) to investigate the effect of fiber hybridization types on mechanical properties. The interface/short beam shear strength, three-point bending strength and tensile strength were obtained experimentally. The full-field strain was detected through the digital image correlation method to reveal the hybrid mechanism. The following conclusions can be drawn based on the testing results and discussion.

1. Compared to core-shell hybrid rod, the mechanical properties of uniform dispersed hybrid rods were 10.9% higher for the short beam shear strength, 60.3% higher for the three-point bending strength and 58.69% higher for the tensile strength, respectively.
2. Full-field strain analysis revealed that the mechanical failure of core-shell hybrid rods was dominated by the interface debonding of the shell/core layers. In comparison, uniform dispersed hybrid rods showed intact

interface bonding between carbon/glass fiber and resin matrix. The tensile strength of the carbon fiber was fully utilized, improving the mechanical properties.

3. The larger scale carbon fiber tow may form the weaker interface layer of carbon/glass fiber/resin, which led to uneven interface stress distribution under the external loading and the decrease of interface shear strength.

Conflict of interest

The authors declare that they have no conflict of interest.

Funding

This research work was funded by National Natural Science Foundation of China (51878223 and 52008137), China Postdoctoral Science Foundation (2019TQ0079 and 2019M661288) and Natural Science Foundation of Heilongjiang Province (LH2021E073).

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